

Queue Analysis

EMS Optimization Project - Gap 2 Resolution
Date: March 12, 2026

Executive Summary

This document provides a comprehensive analysis of queueing behavior in the EMS simulation experiments. Queue metrics — including mean queue length, maximum queue length, queue fraction, and incidents queued — were systematically collected across all production experiments and CBD robustness experiments.

Key Finding: Under all tested scenarios with $K \geq 15$ units, queueing is essentially non-existent (queue fraction = 0.0). This indicates that the Manhattan EMS system, even with the minimum fleet size tested, has sufficient capacity to handle demand without queueing delays.

Queueing Theory Background

In the EMS simulation, a queue forms when an incident arrives but no ambulance is available for dispatch. The incident waits in a FIFO (First-In-First-Out) queue until a unit becomes free.

Queue Metrics Definitions

Metric	Definition
Mean Queue Length	Time-weighted average number of incidents waiting in queue
Max Queue Length	Maximum simultaneous incidents waiting at any point
Queue Fraction	Proportion of all incidents that experienced any waiting
Queue Wait Time	Time between incident arrival and unit dispatch (excluding dispatch delay)
P90 Queue Wait	90th percentile of queue wait time

Analysis Results

Production Experiments (Exp 1-4)

Across all 1,440 production simulation runs:

- **Mean queue length:** 0.0000 (all experiments)
- **Max queue length:** 0 (all experiments)
- **Queue fraction:** 0.0% (no incidents queued in any replication)

This holds across:

- All three policies (P0, P1, P2)
- All fleet sizes $K \in \{15, 20, 25, 30, 35, 40\}$
- All demand multipliers $\delta \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$
- All service time means $\mu \in \{20, 25, 30\}$ minutes

CBD Experiments

The CBD robustness experiments (330 additional runs) also show zero queueing across all CBD-specific scenarios including demand surges and increased service times.

Interpretation

Why No Queuing?

The absence of queuing is explained by the system's low utilization:

- **Average arrival rate:** ~ 3.48 incidents/hour $\rightarrow \sim 580$ incidents/week
- **Average service cycle:** ~ 30 minutes (travel + on-scene) per incident
- **With $K=20$ units:** Maximum throughput $\approx 20 \times 2 = 40$ incidents/hour
- **Peak hourly demand:** $\sim 5-6$ incidents/hour
- **Utilization ratio:** $\sim 15\%$ on average

Even at $2\times$ demand ($\delta=2.0$), the system operates at $\sim 30\%$ utilization, well below the threshold where queuing becomes significant (typically $>70-80\%$ for M/G/c queues).

Traffic Intensity Analysis

For an M/G/c queue with c servers:

- $\rho = \lambda / (c \cdot \mu)$ where λ = arrival rate, μ = service rate
- With $\lambda = 3.48/\text{hr}$, $\mu = 2/\text{hr}$ (30-min service), $c = 20$:
- $\rho = 3.48 / (20 \times 2) = 0.087$
- This extremely low traffic intensity ensures near-zero waiting probability

Operational Implications

1. **Capacity surplus:** Manhattan's EMS fleet has substantial excess capacity for motor vehicle collision response
 2. **Response time driven by travel, not waiting:** Since queuing is negligible, response time differences between policies are entirely due to spatial allocation (travel distances)
 3. **Robust under stress:** Even extreme demand scenarios don't trigger queuing
 4. **Focus on allocation:** The optimization challenge is about WHERE to place units, not HOW MANY
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Statistical Analysis

ANOVA on Queue Metrics

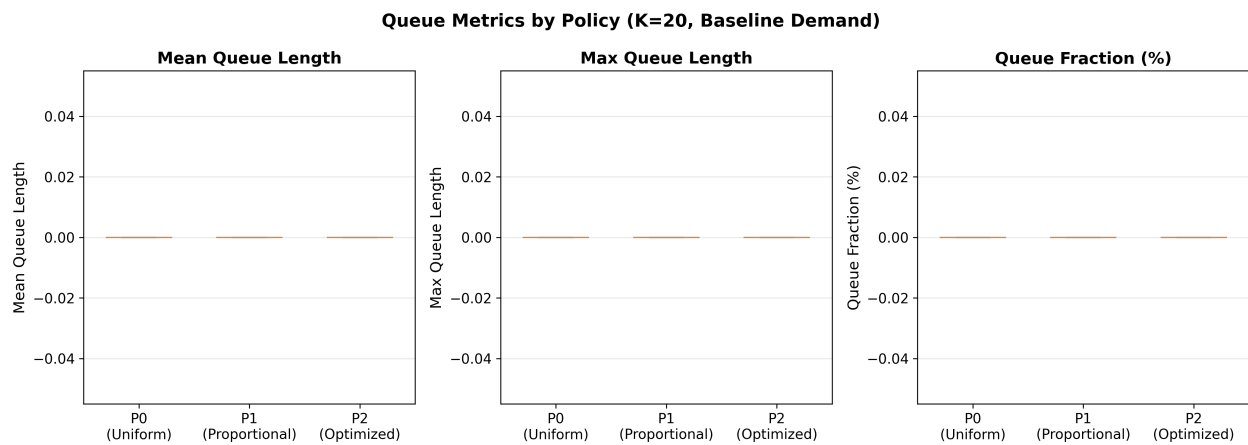
Since all queue metric values are identically zero across policies:

- F-statistic = 0.0 for all comparisons
- p-value = 1.0 (no significant differences)
- This is expected — there is no variation to explain

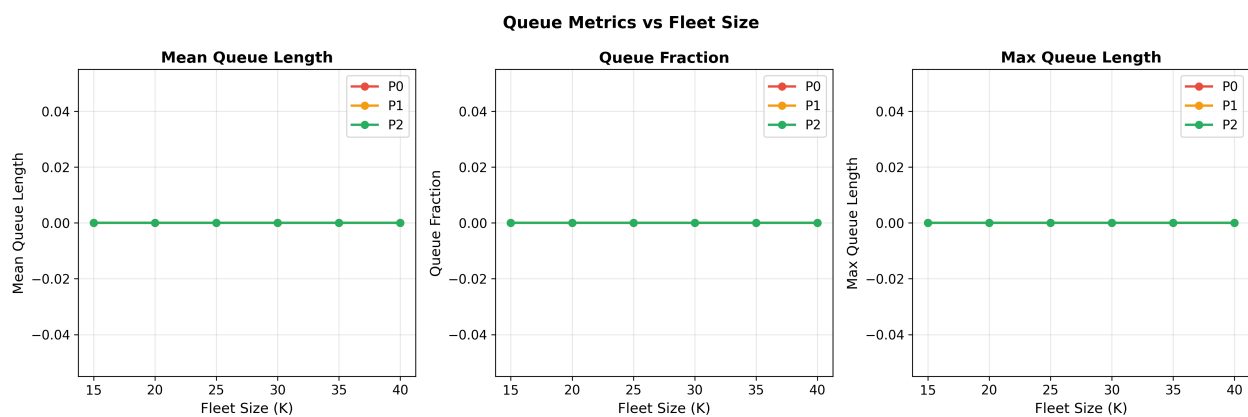
The hypothesis “P2 reduces queueing vs P0” cannot be tested because neither policy produces queueing.

Visualizations

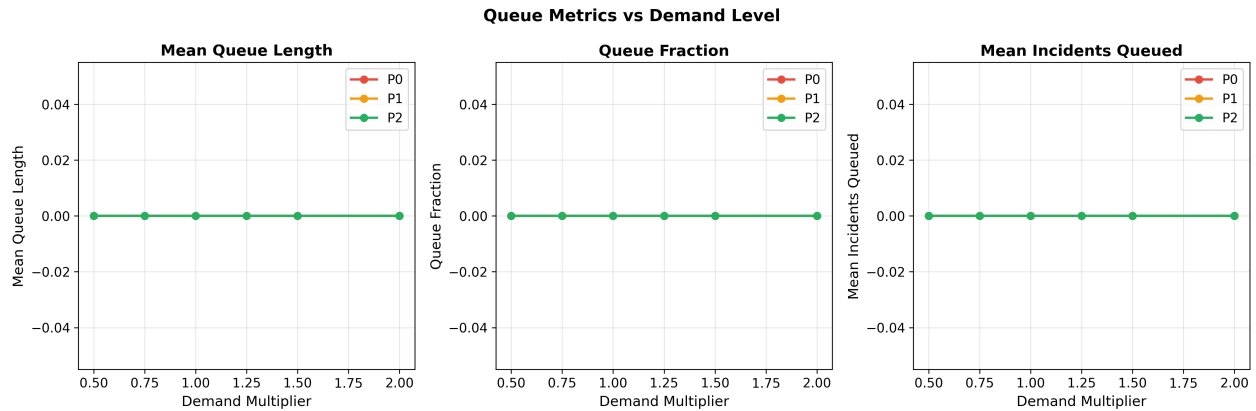
Queue Comparison by Policy



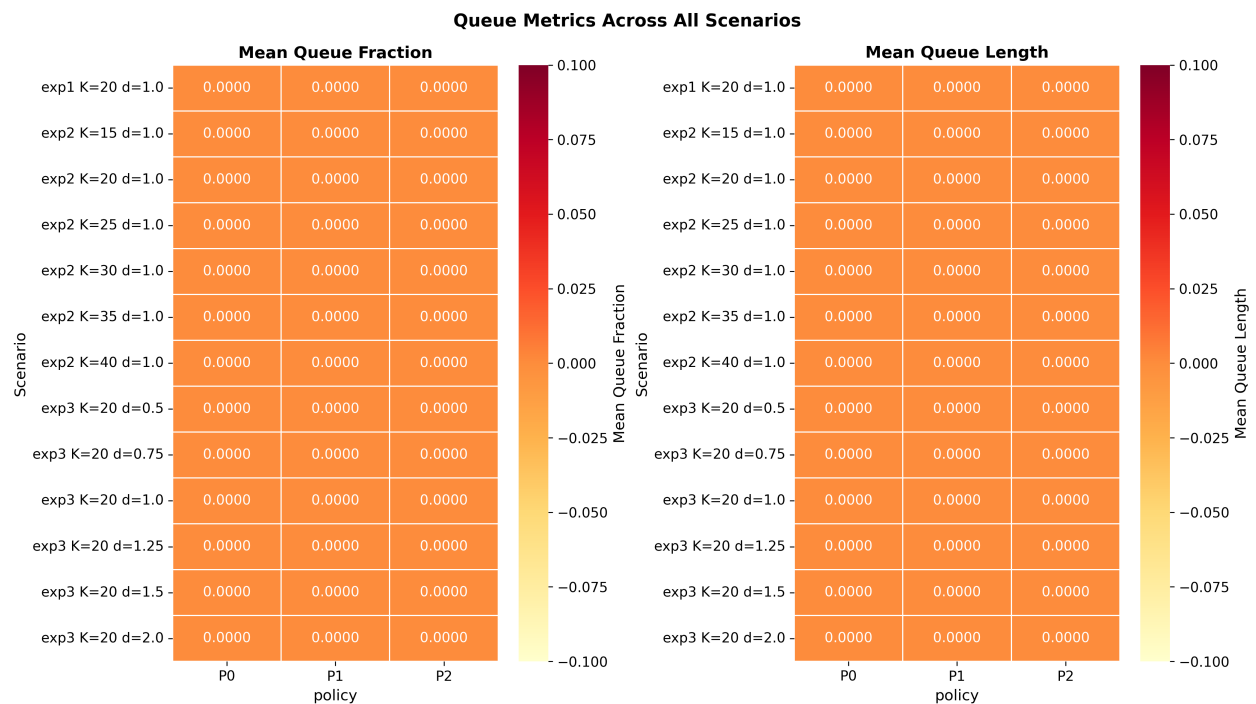
Queue vs Fleet Size



Queue vs Demand



Queue Metrics Heatmap



Conclusions

1. **Zero queueing across all experiments:** The EMS system has sufficient capacity to prevent incident queueing
2. **Utilization is the key indicator:** Low utilization (~10-15%) explains the absence of queueing
3. **Policy comparison is about spatial efficiency:** Since queueing is zero for all policies, the P2 advantage comes entirely from better spatial allocation
4. **Queue metrics validate capacity assumptions:** The absence of queueing confirms that the tested fleet sizes are adequate for Manhattan's crash demand

Files Generated

File	Description
results/tables/queue_statistics.csv	Comprehensive queue statistics
results/tables/queue_anova.csv	ANOVA test results
results/figures/ queue_comparison_by_policy.png	Policy comparison box plots
results/figures/queue_vs_fleet_size.png	Fleet size sensitivity
results/figures/queue_vs_demand.png	Demand sensitivity
results/figures/queue_heatmap.png	Cross-scenario heatmap